

Inventory Tracking System for Non-Profits

Reflection Document

Group 14

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Link to Github Repo

<https://github.com/andrewgoetz/Capstone2025> (Final version of SRS, Project Plan, Design Doc and V&V document, demo video, poster file are in folder Capstone2025/Final Versions Of Documents)

Project Complexity

This project demonstrates a **high level of complexity** due to its nature as a fully integrated, end-to-end system that spans multiple layers of software engineering. As outlined in the SRS, the platform combines a modern full-stack architecture—React frontend, Python-based backend, and PostgreSQL database—with advanced features such as barcode scanning, bulk data ingestion, and real-time analytics. The implementation of these features required careful coordination between frontend interfaces, backend APIs, and data storage systems to ensure consistency, performance, and usability. Additionally, the system handles multiple forms of data input (manual entry, barcode scans, CSV uploads) and transforms them into structured, queryable data, demonstrating significant complexity in both system design and data engineering.

The project's complexity is further elevated by the integration of a machine learning-based demand forecasting component into a production-like environment. As specified in the SRS, this involves building a full data pipeline, training models on historical data, and evaluating performance using metrics such as WAPE and forecasting bias. In practice, this required overcoming real-world challenges such as acquiring reliable datasets over sufficient periods of time and ensuring that predictions meaningfully inform system features like stock planning and dashboards. Combined with additional requirements such as role-based authentication, security constraints, real-time reporting, and cross-platform development through a mobile app, the system moves beyond a simple application into a cohesive, scalable platform. The incorporation of real stakeholder feedback from McMaster's campus food bank – the *Food Collective Centre (FCC)* – and the system's potential for real-world deployment further reinforce its complexity, as it must satisfy both technical requirements and practical usability constraints.

Project Achievements

Our project delivers a fully functional, end-to-end inventory management platform tailored to the operational needs of non-profit organizations. A key achievement was the implementation of a complete barcode-driven workflow that supports both item intake and distribution through structured scan-in and scan-out processes. Our system allows users to process inventory either individually, through single-item scans, or in bulk via a dedicated check-in/check-out interface, enabling efficient handling of both low- and high-volume donation environments. Additionally, it also supports bulk data ingestion through CSV/Excel uploads and provides a searchable, filterable interface for managing inventory. These features significantly improve usability compared to traditional manual or spreadsheet-based approaches.

Another major achievement was the integration of data analytics and reporting capabilities directly into the platform. The dashboard provides real-time insights into inventory distribution, low-stock alerts and expiring items, allowing organizations to make quicker and more informed operational decisions. The system also captures detailed item attributes, including dietary restrictions (e.g., Kosher, Halal, etc.) and storage locations, enabling volunteers to quickly identify appropriate items

and make more informed, need-based distribution decisions for different families. This adds a layer of practicality to the system beyond basic inventory tracking.

Our project also integrates a machine learning-based demand forecasting component. We leveraged historical inventory movement data acquired from an actual food bank to generate short-term demand predictions, enabling organizations to proactively plan resource allocation, anticipate shortages and better manage incoming donations.

In addition to technical functionality, our project demonstrates strong real-world applicability and accessibility. This platform was intentionally designed as a free, open-source solution, making it a cost-effective alternative to expensive proprietary systems commonly used by non-profits. This significantly lowers the barrier to adoption for organizations with limited resources. The system was also iteratively refined using feedback from the volunteers and executives at the FCC ensuring alignment with real operational workflows and usability needs. Lastly, the FCC has expressed interest in adopting the system upon completion, highlighting the project's potential for real-world deployment and tangible community impact.

Completion of Requirements

Requirement - Priority Level	Expected Features
Inventory Management - P0	Barcode Scanning UI, Lookup and Registration Item Management Interface Location Selector Bulk Upload & Import Handler Inventory Schema & Upload
Dashboard - P0	Graphical Reports & Widgets Export Options Aggregate Queries Scheduled Jobs
Documentation - P0	Setup Guide User Manual Unit Testing
Demand Forecasting - P1	Forecast Dashboard Visualization Charts Machine Learning Model

	Data Pipeline Stock vs. Demand Algorithm
Role-Based Access/Authentication - P1	Login & Registration Role-Based Interface Authentication Endpoints Roles & User Tables
Total Requirement Completion Percentage	21/21 = 100%

Highlights and Lowlights

One of the most meaningful highlights came from outside our own team. During the building of our demand forecasting model, we developed an ongoing relationship with McMaster's FCC, getting their feedback on usability and overall functionality. Their response was very enthusiastic, as they noted that our system would meaningfully replace the manual, fragmented processes they currently rely on and expressed genuine interest in adopting the platform. Receiving that kind of validation from a real primary stakeholder with real operational needs was a strong signal that we had built something with tangible, lasting impact.

Another standout highlight we achieved was the successful implementation of our mobile application. Going into the project, the mobile app was scoped as a P2 stretch feature, as we were uncertain whether we would have enough time to build it properly, and no one on the team had prior mobile development experience. Despite that, the implementation went smoothly. By splitting the team effectively, a few members were able to focus on the mobile app in parallel while the rest of the system continued to be worked on, avoiding any bottlenecks. What we initially treated as a "nice to have" feature ended up as a polished, fully integrated component of the final product.

The most significant lowlight moment we faced was acquiring sufficient, real-world data for our demand forecasting model. Early in development, we recognized that meaningful forecasting required historical inventory distribution data from actual food banks, not just a public dataset. We reached out to multiple local food banks to explore potential data-sharing arrangements, but for an extended period, none were willing to provide their data (e.g. due to privacy concerns, organizational policies, or simply a lack of historical outbound tracking data). Simultaneously, our search for open-source alternatives was not successful, as the publicly available datasets we found lacked the volume and specificity needed to identify meaningful trends or seasonal patterns relevant to our use case. This created a period of uncertainty around one of our P1 features, and forced us to place the forecasting component on hold while we worked on other parts of the system. The turning point came when the FCC responded positively to our outreach and agreed to share their inventory distribution records with us. With access to real operational data, we were able to retain and refine our model in a way that reflected genuine demand patterns, improving the relevance and reliability of our forecasting output.

What We Learnt

One of the biggest takeaways from this project was learning what it really means to build software for real users. At the beginning, many of our decisions were driven directly by the SRS, but once we started working with the Food Collective Centre, our perspective shifted. Talking to actual users highlighted priorities that no specification can fully capture, like how volunteers naturally move through a workflow, what information they need immediately, and the small pain points that can make or break usability. The FCC's feedback pushed us to focus less on simply completing features and more on creating something that felt practical and intuitive in a real environment.

From a technical perspective, accurate demand forecasting for food banks requires a minimum of two years of consistent distribution history to capture annual seasonality; but with sparse or short real-world data (like McMaster food bank's), simpler intermittent-demand models such as Croston TSB outperform more sophisticated seasonal methods like Holt-Winters ETS. Also, none of us had prior experience with mobile development, so building the mobile app was a new challenge. We learned how to use Expo and React Native to build a cross-platform application with features like inventory browsing, checkout, and activity tracking, and gained an understanding for how mobile development differs from web, particularly around navigation patterns and touch interactions.

Finally, coordinating a team of seven required far more intentional organization than previous coursework. Dividing into parallel workstreams and treating tasks like focused mini-sprints helped us stay productive and adapt to changing priorities.